Binomial random variables

1 In a binomial experiment:

a Are the trials independent or dependent?

b How many possible outcomes are there for each trial?

2 State whether following scenarios describe a binomial random variable:

a Rolling a six-sided die 20 times and counting the number of times the die lands on 1.

b The sum of the outcomes of rolling a six-sided die 10 times.

c Selecting 70 random people from the population and recording the number of females.

d A particular coin is more likely to land on heads than tails. Tossing this coin 10 times and recording the number of times it lands on heads.

e Drawing a marble with replacement from a bag containing purple, green and yellow marbles, and noting the number of purple marbles.

f Drawing a marble with replacement from a bag containing purple, green and yellow marbles, and noting the number of red marbles.

3 Consider the binomial probability formula:

$$P(x) = {}^{n}C_{x}p^{x}(1-p)^{n-x}$$

where $P(x)$ is the probability of obtaining x successes in n independent trials.

a What does p represent in the formula?

b What does $(1-p)$ represent in the formula?

4 For a binomial probability distribution with n, trials and probability of success p, Ned performed the following calculation:

$$\binom{n}{2}p^2 \times 0.7^4$$

a State the probability of success.

b State the number of trials.

c What probability is Ned trying to find with this calculation?

5 Consider the binomial power $(p+q)^4$.

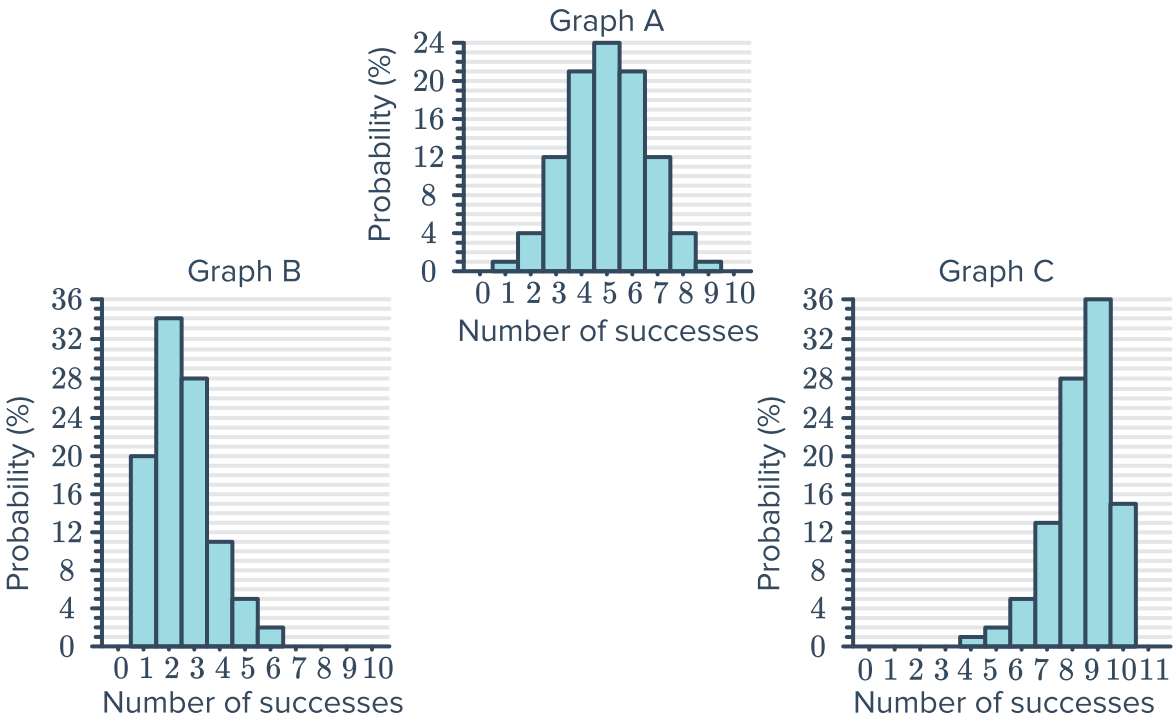
a Expand $(p+q)^4$.

b We can interpret the terms in the expansion in terms of probabilities in a binomial distribution, with p the probability of success and q the probability of failure. What probability would the term $\binom{4}{2}p^2q^2$ in the expansion represent?

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Distributions of binomial random variables

6 Consider the three graphs below that show binomial distributions for $n = 10$ and $p = 0.15$, 0.5 and 0.85:



a Which value of p corresponds to Graph A?

b Which value of p corresponds to Graph B?

c Which value of p corresponds to Graph C?

7 Consider a binomial distribution of a random variable X with $p = 0.5$ and $n = 17$.

a Is $P(\mathrm{X} = 5)$ greater than, less than or equal to $P(\mathrm{X} = 12)$?

b Is the shape of the graph of this binomial distribution symmetric, positively skewed or negatively skewed?

8 Consider a binomial distribution of a random variable X with $p = 0.13$ and $n = 11$.

a Is $P(\mathrm{X} = 4)$ greater than, less than or equal to $P(\mathrm{X} = 7)$?

b Is the shape of the graph of this binomial distribution symmetric, positively skewed or negatively skewed?

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9 Consider a binomial distribution of a random variable X with $p = 0.83$ and $n = 17$.

a Is $P(\mathrm{X} = 2)$ greater than, less than or equal to $P(\mathrm{X} = 15)$?

b Is the shape of the graph of this binomial distribution symmetric, positively skewed or negatively skewed?

10 For each of the followng table of values for the binomial distribution X:

i Describe the shape of the graph as positively skewed, symmetric, or negatively skewed.

ii Calculate the value of p correct to two decimal places.

a

x	0	1	2	3	4	5	6	7	8
$P(X = x)$	0.004	0.031	0.109	0.219	0.274	0.219	0.109	0.031	0.004

b

x	0	1	2	3	4	5	6	7	8
$P(X = x)$	0.152	0.323	0.3	0.16	0.053	0.011	0.001	0	0

c

x	0	1	2	3	4	5	6	7	8
$P(X = x)$	0	0	0.001	0.009	0.046	0.147	0.294	0.336	0.168

Probability for binomial random variables

11 Find the value of the following:

a ${}^5C_4 \times (0.1)^4 \times 0.9 + {}^5C_5 \times (0.1)^5 \times (0.9)^0$

b ${}^5C_3 \times (0.8)^3 \times (0.2)^2 + {}^5C_4 \times (0.8)^4 \times 0.2 + {}^5C_5 \times (0.8)^5 \times (0.2)^0$

12 A binomial random variable X has 13 independent trials and a probability of success of 0.4.

a Using the definition for the binomial random variable X, write the formula for $P(X = 2)$.

b Calculate $P(X = 2)$ to four decimal places.

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13 For a binomial random variable X, $P(X=3) = {}^{10}C_3 \times 0.6^3 \times 0.4^7$.

a Calculate $P(X=3)$ to four decimal places.

b State the number of trials in this experiment.

c State the probability of success in this experiment.

14 X is a binomial variable with the probability mass function:

$$P(X=k) = {}^4C_k \times (0.4)^k \times (0.6)^{4-k} \text{ for } k = 0,1,2,3,4$$

a State the number of trials for this distribution.

b State the probability of success.

c Wrie an expression for $P(X=2)$.

d How many ways can we get 2 successes in the 4 trials?

e Calculate the probability $P(X=2)$.

15 X is a binomial variable with the probability mass function:

$$P(X=k) = {}^3C_k \times (0.3)^k \times (0.7)^{3-k} \text{ for } k = 0,1,2,3$$

Complete the table:

x	0	1	2	3
$P(X=x)$	0.343	0.441		

16 Consider the random variable X, which has distribution $X \sim B(4, 0.1)$.

a Write an expression for $P(X=1)$.

b Write an expression for $P(X=2)$.

c Complete the following table:

x	0	1	2	3	4
$P(X=x)$	0.6561				0.0001

17 A binomial random variable X is defined as:

$$P(X=x) = {}^{20}C_x \times 0.75^x \times 0.25^{20-x} \text{ for } x = 1,2,\ldots,20$$

a State the number of trials in this experiment.

b State the probability of success in this experiment.

c Calculate $P(X=11)$ to four decimal places.

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$$P(X = k) = {}^{6}C_k \times (0.8)^k \times (0.2)^{6-k} \text{ for } k = 0, 1, 2, 3, 4, 5, 6$$

a State the number of trials for this distribution.

b What is the probability of failure?

c Calculate the probability $P(X = 3)$.

d Calculate the probability $P(X \leq 3)$.

19 X is a binomial variable with the probability mass function:

$$P(X = k) = {}^{5}C_k \times (0.4)^k \times (0.6)^{5-k} \text{ for } k = 0, 1, 2, 3, 4, 5$$

a Calculate $P(X \geq 3)$.

b Calculate $P(X < 3)$.

20 X is a binomial variable with the probability mass function:

$$P(X = k) = {}^{5}C_k \times (0.2)^k \times (0.8)^{5-k} \text{ for } k = 0, 1, 2, 3, 4, 5$$

a State the number of trials for this distribution.

b State the probability of success.

c Calculate the probability $P(X > 3)$.

d What is the most likely number of successes?

e Calculate the mean of the distribution.

f Calculate the variance of the distribution.

21 Consider the random variable X, which has distribution $X \sim B\left(8, \frac{4}{5}\right)$.

a State the number of trials for this distribution.

b State the probability of success.

c Calculate the probability $P(X > 4)$ to two decimal places.

d What is the most likely number of successes?

e Calculate the mean of the distribution.

f Calculate the variance of the distribution.

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22 X is a binomial variable with a probability distribution given in the table:

x	0	1	2	3	4	5
$P(X = x)$	0.00243	0.02835	0.1323	0.3087	0.36015	0.16807

a Find $P(X \geq 4)$.

b Find $P(X < 3)$.

23 The following table represents a probability distribution of a binomial random variable:

x	0	1	2	3	4
$P(X = x)$	0.2401	0.4116	0.2646	0.0756	0.0081

a State the number of trials for this distribution.

b Calculate the probability $P(X = 3)$.

c Calculate the probability $P(X < 3)$.

d Using $P(X = 4)$, calculate the probability of success p.

e Calculate $E(X)$.

f Calculate the standard deviation of the distribution.

24 The table represents a cumulative probability distribution of a binomial random variable:

x	0	1	2	3	4	5
$P(X \leq x)$	0.01024	0.08704	0.31744	0.66304	0.92224	1

a State the number of trials for this distribution.

b Calculate $P(X \leq 2)$.

c Calculate $P(X \geq 3)$.

d Calculate $P(X = 5)$.

e Using $P(X = 5)$, calculate the probability of success p.

f Calculate the standard deviation of the distribution.

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25 The following graph represents a probability distribution of a binomial random variable:

a State the number of trials for this distribution.

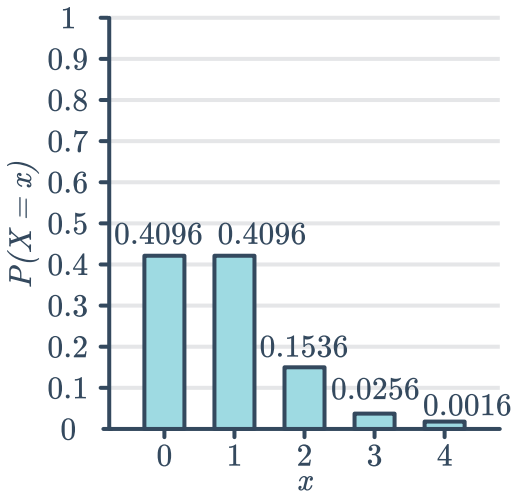
b Calculate $P(X = 3)$.

c Calculate $P(X < 3)$.

d Using $P(X = 4)$, calculate the probability of success p.

e Calculate $E(X)$.

f Calculate the standard deviation of the distribution.



26 The following graph represents a cumulative probability distribution of a binomial random variable:

a State the number of trials for this distribution.

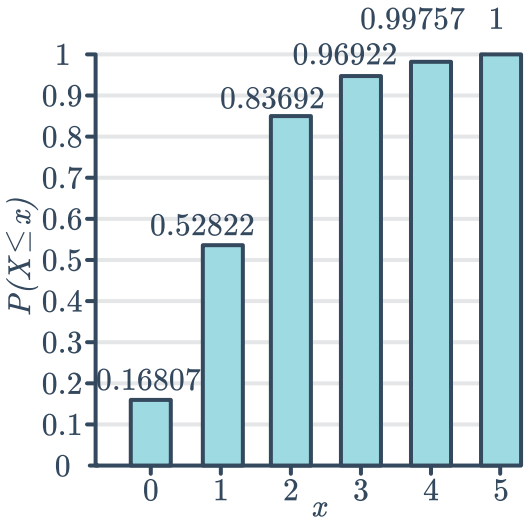
b Calculate $P(X \leq 2)$.

c Calculate $P(X \geq 3)$.

d Calculate $P(X = 5)$.

e Using $P(X = 5)$, calculate the probability of success p.

f Calculate the standard deviation of the distribution.



27 Calculate the probability of the following, rounding your answers to two decimal places:

a Getting exactly 18 successes in 30 independent trials if the probability of success in each trial is 0.7.

b Getting at least 10 successes in 80 independent trials if the probability of success in each trial is $p = 0.1$.

c Getting at most 22 successes in 25 independent trials if the probability of success in each trial is 0.85.

d Getting more than 15 successes in 33 independent trials if the probability of success in each trial is 0.45.

e Getting less than 12 successes in 50 independent trials if the probability of success in each trial is 0.28.

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28 $P(X = 2) = 0.3456$ for a binomial random variable X with 4 trials.

a If success is more likely, find the probability of success p.

b Calculate $E(X)$.

c Calculate $\text{Var}(X)$.

29 $E(X) = 6$ for a binomial random variable X with 10 trials.

a Find the probability of success p.

b Calculate $\text{Var}(X)$.

30 The expected value of a binomial distribution is 8 and the variance is 6.4.

a Find the number of trials, n.

b Find the probability of success, p.

31 $E(X) = \dfrac{15}{2}$ for a binomial random variable X with probability of success $\dfrac{3}{10}$.

a Find the number of trials n.

b Calculate the standard deviation of X.

32 $E(X) = 2$ and $\text{Var}(X) = 1.6$ for a binomial random variable X.

a Find the probability of success p.

b Find the number of trials n.

33 $E(X) = 4.8$ and $\sigma(X) = \sqrt{2.88}$ for a binomial random variable X.

a Find the probability of success p.

b Find the number of trials n.

Applications

34 A certain disease has a survival rate of 64%. Of the next 110 people who contract the disease, how many would you expect to survive? Round your answer to the nearest whole number.

35 The probability of a particular tennis player getting their first serve in is 0.72. If she serves 90 times in a match, how many times would she expect to get her first serve in? Round your answer rounded to the nearest whole number.

36 A fair die is rolled 5 times and the number of fours noted.

a On each trial, what is the probability of success?

b Calculate the probability of rolling exactly 2 fours, correct to four decimal places.

c Calculate the probability of rolling exactly 3 fours, correct to four decimal places.

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37 David owns 3 yellow mugs and 8 blue mugs. Each morning, he chooses a mug at random for his cup of tea before work, 5 days a week. The mug is washed and returned before the next day.

a Each day, what is the probability of choosing a yellow mug?

b What is the probability that he used a yellow mug exactly twice this past work week? Round your answer to four decimal places.

c What is the probability that he never used a yellow mug this past work week? Round your answer to four decimal places.

38 At a local Italian restaurant, 16 people are in line to order. The probability that any one person will choose pizza is 23%.

a What is the probability that exactly 4 people choose pizza? Round your answer to four decimal places.

b What is the probability that exactly half of the people choose pizza? Round your answer to four decimal places.

39 Consider the binomial $\left(\frac{1}{5}+\frac{4}{5}\right)^5$.

a Write the formula for the term of the expansion that contains $\left(\frac{4}{5}\right)^3$.

b A box of chocolates is $\frac{1}{5}$ dark chocolate and $\frac{4}{5}$ milk chocolate. If 5 chocolates are selected at random with replacement, describe what the term from part (a) would tell us.

c Calculate the probability of selecting exactly 1 dark chocolate.

d Calculate the probability of selecting at most 1 dark chocolate.

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